

DATA CREATION

Objective:

Optimizing Travel Package Performance, Customer Satisfaction Analysis & Data Collection enhancement.

Tools & Environment:

1. IDE: Visual Studio Code (VS Code)
2. Language: Python
3. Libraries Used: pandas, numpy, random, datetime, uuid

Methodology:

The data was created using custom Python scripts with the following structure:

1. Define schema: Identified columns and data types for the dataset (e.g., customer name, product, sales amount, transaction date).
2. Generate synthetic data: Used loops and libraries like random and numpy to generate realistic values.
3. Introduce variability: Added randomness to simulate real business data (e.g., seasonal sales spikes, missing values).
4. Save output: Exported the final dataset as a .csv file for ingestion into GCS.

Limitations:

1. Data is synthetic and not reflect the original values related to company performance
2. Data only available based on what authors decided to include
3. Some parts cannot be implemented due to data limitations and time constraints

1. Customers Table:

a) Sample codes:

https://github.com/nasrulkhair/Full-Project/blob/main/PPNJ/data_creation/data.ipynb

b) Output Description:

- File created: customers.csv
- Rows:
- Columns:

c) Table Schema

Column Name	Type	Description
customer_id	STRING	NON_NULLABLE
customer_name	STRING	NULLABLE
gender	STRING	NULLABLE
age	INTEGRER	NULLABLE
email	STRING	NULLABLE
customer_type	STRING	NULLABLE

2. Travel Package Table

a) Sample codes:

https://github.com/nasrulkhair/Full-Project/blob/main/PPNJ/data_creation/data.ipynb

b) Output Description:

- File created: travel_package.csv
- Rows:
- Columns:

c) Table Schema:

Column Name	Type	Description
package_id	STRING	NON_NULLABLE
package_name	STRING	NULLABLE
package_type	STRING	NULLABLE
duration_days	INTEGER	NULLABLE
price_per_pax	INTEGER	NULLABLE
category	STRING	NULLABLE
destination	STRING	NULLABLE

3. Travel Guide Table

a) Sample codes:

https://github.com/nasrulkhair/Full-Project/blob/main/PPNJ/data_creation/data.ipynb

b) Output Description:

- File created: travel_guide.csv
- Rows:
- Columns:

c) Table Schema:

Column Name	Type	Description
staff_id	INTEGER	NON_NULLABLE
staff_name	STRING	NULLABLE
role	STRING	NULLABLE
working_status	INTEGER	NULLABLE
comission_rate	FLOAT	NULLABLE

4. Sales Table

a) Sample codes:

https://github.com/nasrulkhair/Full-Project/blob/main/PPNJ/data_creation/data.ipynb

b) Output Description:

- File created: sales.csv
- Rows:
- Columns:

c) Table Schema:

Column Name	Type	Description
booking_id	STRING	NON_NULLABLE
customer_name	STRING	NULLABLE
package_id	STRING	NULLABLE
guide_id	INTEGER	NULLABLE
booking_date	DATE	NULLABLE
travel_date	DATE	NULLABLE
num_of_pax	INTEGER	NULLABLE
discount_promotion	INTEGER	NULLABLE
channel	STRING	NULLABLE
payment_status	STRING	NULLABLE
payment_method	STRING	NULLABLE

5. Feedback Table

a) Sample codes:

https://github.com/nasrulkhair/Full-Project/blob/main/PPNJ/data_creation/data.ipynb

b) Output Description:

- File created: feedback.csv
- Rows:
- Columns:

c) Table Schema:

Column Name	Type	Description
feedback_id	STRING	NON_NULLABLE
package_id	STRING	NULLABLE
booking_id	STRING	NULLABLE
satisfaction_score	INTEGER	NULLABLE
feedback_date	DATE	NULLABLE
comments	STRING	NULLABLE
rating_hotel	INTEGER	NULLABLE
rating_guide	INTEGER	NULLABLE
rating_itinerary	INTEGER	NULLABLE
would_recommend	STRING	NULLABLE

6. Refund Table

d) Sample codes:

https://github.com/nasrulkhair/Full-Project/blob/main/PPNJ/data_creation/data.ipynb

e) Output Description:

- File created: refund.csv
- Rows:
- Columns:

f) Table Schema:

Column Name	Type	Description
refund_id	STRING	NON_NULLABLE
booking_id	STRING	NULLABLE
customer_id	STRING	NULLABLE
refund_date	DATE	NULLABLE
refund_amount	INTEGER	NULLABLE
refund_percentage	FLOAT	NULLABLE
reason	STRING	NULLABLE
refund_status	STRING	NULLABLE

Transformations:

https://github.com/nasrulkhair/Full-Project/blob/main/PPNJ/data_creation/data.ipynb

No.	Table	Column	Transformation
1.	customers	email	fill null values with 'not provided'
2.	customers	email	replacement: , → " / ' : ' → ' '
3.	feedback	feedback_date	convert data type from STRING to DATE
4.	refund	refund_date	convert data type from STRING to DATE
5.	travel_guide	role	' ' → ''
6.	sales	payment_method	replace ' ' → ''